

App Dev Project Report

1. Student Details

Field	Details
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2. Project Details

Project Title: HimTrek – Trekking Management Application

Problem Statement:

Adventure organizations rely on manual coordination (spreadsheets, calls) to manage trekking activities, leading to overbooking, missed reminders, and poor visibility into participant data. This project builds a web application that brings Admins, Trek Staff, and Trekkers onto one structured platform with clear role boundaries.

Approach:

The app was built using Flask as the backend API with JWT-based authentication and role-based route decorators. VueJS 3 serves as the SPA frontend with Pinia for state management and Vue Router for client-side navigation guards. Background tasks (daily reminders, monthly reports, CSV exports) are handled asynchronously using Celery with Redis as the message broker. An outbox pattern is implemented to ensure tasks are not lost if the worker is temporarily down.

3. AI/LLM Declaration

I used AI assistance (Claude / Gemini) for approximately **25%** of the project, specifically for:

- Repetitive boilerplate code** – generating the initial structure of similar Flask Blueprint files (e.g., `staff.py`, `admin.py`) once the first Blueprint pattern was established.
- Documentation formatting** – structuring markdown files (README, this report) and inline code comments.

5. Database Schema



6. API Resource Endpoints

Auth

Endpoint	Method	Role	Description
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Endpoint	Method	Role	Description
/api/auth/register	POST	Public	Register new trekker account
/api/auth/login	POST	Public	Login and receive JWT token
/api/auth/me	GET	Any	Get own profile
/api/auth/me	PUT	Any	Update profile / change password (requires current password)

Treks

Endpoint	Method	Role	Description
/api/treks/	GET	Public	List all open treks
/api/treks/search	GET	Public	Filter treks by location, difficulty, duration
/api/treks/	POST	Admin	Create new trek
/api/treks/<id>	PUT	Admin/Staff	Update trek metadata or status
/api/treks/<id>	DELETE	Admin	Delete a trek
/api/treks/<id>/assign	POST	Admin	Assign a staff member to a trek

Bookings

Endpoint	Method	Role	Description
/api/bookings/	POST	Trekker	Book a trek (slot check included)
/api/bookings/	GET	Trekker	Get own booking history
/api/bookings/<id>/cancel	PUT	Trekker/Admin	Cancel a booking
/api/bookings/export	GET	Trekker	Queue async CSV email export

Admin

Endpoint	Method	Role	Description
/api/admin/dashboard	GET	Admin	Platform stats (users, treks, bookings)
/api/admin/users	GET	Admin	List trekkers or staff with search

Drive Link:

<https://drive.google.com/file/d/1nXQOTScg9CfIgrKzgWpYZwwFTf76tvWl/view?usp=sharing>